Global Alignment Practice Questions - Answers

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Question 1: Global alignment (Needleman-Wunsch)

Seq 1: G A C T Seq 2: G T C T

Scoring scheme: Match = +1 Mismatch = -1 Gap = -3

Matrix:

	-	G	T	C	T
-	0	-3	-6	-9	-12
G	-3	**1**	-2	-5	-8
A	-6	-2	**0**	-3	-6
C	-9	-5	-3	**1**	-2
T	-12	-8	-4	-2	**2**

Final alignment:

G A C T
G T C T

Total alignment score: 2

Question 2: Global alignment (Needleman-Wunsch)

Seq 1: T A C G Seq 2: T A C C G

Scoring scheme: Match = +1 Mismatch = -1 Gap = -3

Matrix:

	-	T	A	C	C	G
-	0	-2	-4	-6	-8	-10
T	-2	**2**	0	-2	-4	-6

A	-4	0	**4**	2	0	-2
C	-6	-2	2	**6**	4	2
G	-8	-4	0	4	5	**6**

Final alignment:

Possible Alignment 1:

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T A C C G
T A - C G
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Possible Alignment 2:

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T A C C G
T A C - G
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Total alignment score: 6